

SAGAR MANDIYA

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EDUCATION

Boston University

Master of Science in Computer Science

Relevant Coursework: Artificial Intelligence, Machine Learning, Software Engineering, Analysis of Algorithms, Data Science with Python, Web Mining and Analytics

Boston, MA

Expected Jan 2024

SKILLS & CERTIFICATION

Technical Languages: Python, C, C++, R, MATLAB, SQL, JavaScript, HTML, CSS, Lua

Frameworks: TensorFlow, PyTorch, CUDA, TensorRT, Nvidia Riva, Nvidia Triton, Nvidia Deepstream, FastAPI, Django

Developer Tools: Git, Docker, WordPress, Bootstrap, Oracle Cloud Platform, Latex, VS Code, PyCharm, IntelliJ

Libraries: Pandas, NumPy, Matplotlib, OpenCV, Scikit-learn, SciPy/nltk

Operating Systems: Ubuntu, Kali, Fedora, MacOS, Windows

Certifications: Deep Learning Specialization, Convolutional Neural Networks in TensorFlow, Machine Learning

Extra-curriculars: Soccer, Swimming, Video Games

EXPERIENCE

Graduate Research Assistant

Boston University - Department of Computer Science

Boston, MA

Apr 2023 – Present

- Analyzed ER (Emergency Room) data using data science and machine learning techniques, deriving insights to minimize machinery-related work injuries to an estimate of more than 20%.
- Generated compelling data visualizations using Python and Tableau, reducing injury misclassification by 15% and enabling more accurate injury severity and recovery time predictions.
- Conducted data analysis for groundbreaking autism study, clarifying optimal age for improvement saturation in autistic children.
- Leveraged rigorous data visualization techniques alongside machine learning, uncovering key trends in a vast autism dataset, leading to a 25% increase in data-driven insights.

Boston University - BitLab @ Questrom School of Business

Mar 2023 – Present

- Investigated role of Generative AI in music generation, determining its impact on holistic content creation in current market.
- Constructed LLM (Large Language Models) and transformer pipelines for innovative music generation and content creation.
- Organized thorough testing and interpretation of data, yielding valuable insights potentially guiding the integration of generative AI in music industry landscape.

Computer Vision Engineer

TensorGo Technologies

May 2022 – Aug 2022

- Led a team as a project manager to design and built effective solution for virtual perimeter overlay, utilizing image segmentation and object detection to replace soccer field boundary advertisements.
- Increased product performance by 20% through comprehensive model tuning, optimizing image-related solutions.
- Executed innovative image processing techniques to optimize virtual perimeter overlay, attaining a 30% reduction in processing time while maintaining accuracy.

Associate Computer Vision Engineer

TensorGo Technologies

Sep 2021 – May 2022

- Deployed a comprehensive pipeline for an Answer Verification System (AVS) from scratch using BERT language model, reducing hiring time for recruiters by an estimate of 25%.
- Built a complete pipeline to facilitate end-to-end ASR (Automatic Speech Recognition), Voice Biometrics, and Diarization utilizing Nvidia NGC stack (RIVA, NEMO, Triton), resulting in reduction of inference time by 20%.
- Formulated and managed project collaboratively, addressing issues, and strategically engaging with team members and stakeholders.

PROJECTS

Deepfake Detection | Python, TensorFlow, TensorRT, XceptionNet, MisoNet, ResNet, VGG-19

- Designed and implemented neural network models for detecting artificially generated fake images/videos with 78% accuracy.
- Conducted a comparative study between neural network models to determine performance for deepfake detection. [Paper - <https://arxiv.org/abs/2106.12605>].

Sentiment Analysis from Speech | Python, TensorFlow, LSTM, ASR

- Designed and integrated an LSTM based model capable of detecting sentiments with an accuracy of 97%.
- Paired sentiment model with ASR technology to create a seamless end-to-end speech transcription workflow.